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Estimating Heterogeneous Price Elasticities for Newly Launched Fashion Products via Bayesian Hierarchical Models

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Executive Summary

Fast fashion, a business model that prioritises high product turnover, makes informed pricing decisions challenging as many products have limited sales history. A key metric for making pricing decisions is price elasticity, which quantifies how demand responds to changes in price. Our goal is to improve pricing decisions for newly launched fast fashion products by providing reliable estimates of price elasticity despite limited historical sales data.

To achieve this, we fit a series of Bayesian models to approximately [REDACTED] monthly sales observations across [REDACTED] products from a [REDACTED] fashion retailer. These models incorporate product and category-level effects to capture price sensitivities across individual products and product categories (e.g., bags or shoes), allowing information from products with sales history to inform elasticity estimates for new products while also quantifying the uncertainty of those estimates via posterior credible intervals.

Using these models, we found that [REDACTED]
[REDACTED]
[REDACTED]. Additionally, before accounting for differences in price sensitivity across products and product categories, price elasticity was estimated to be [REDACTED]
[REDACTED]

We therefore recommend [REDACTED]
[REDACTED]

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1 Introduction

The term fast fashion describes products that are affordable, trendy, and rapidly produced. With increasing popularity of fast fashion among retailers, competitors face pressure to optimise pricing decisions in real time. A key metric for making pricing decisions is price elasticity, which quantifies how responsive customer demand is to changes in price. Being able to accurately estimate the price elasticity of a product allows a company to adjust prices to maximise revenue, minimise excessive inventory, and respond to competitor pricing strategies.

Historically, price elasticity has been modeled using pooled methods such as Ordinary Least Squares and Quantile Regression, which estimate a single elasticity across all product groups, without capturing the variation in elasticities between these products [1]. This has led to the idea of applying shrinkage estimation and Bayesian hierarchical models, allowing products to borrow information from similar products to produce more reliable estimates [2], which is particularly valuable in the fast fashion setting, where rapid product turnover leaves many products with limited sales history.

In this report, we aim to estimate and predict price elasticities for newly launched fast fashion products at Simon-Kucher, fitting multiple Bayesian hierarchical models of increasing complexity to assess the benefit of incorporating product and category-level heterogeneity into price elasticity estimation, with uncertainty quantified via posterior credible intervals.

2 Materials and Methods

2.1 Data Overview and Preprocessing

Daily sales data spanning January 2023 to September 2025 were obtained from a [REDACTED] fashion retailer, comprising three tables: a sales table of invoice-level transactions, a product table containing category and season information, and a date table with special event flags, totalling approximately [REDACTED] transactions.

Preprocessing began by joining the date table to the sales table, then removing the [REDACTED] transactions occurring during a non-recurring disruption week, as this pattern cannot generalise to future products. Daily special event flags (detailed in Appendix A) were consolidated into monthly indicators, assigning a value of 1 to any month containing at least one flagged day, capturing broader demand shifts instead of pinpointing specific days.

Data were then grouped by product identifier, month, and year, with quantity sold aggregated as a monthly sum and sales price as a monthly median, as the median is robust to price outliers. Monthly aggregation was preferred over invoice-level modelling as it creates variance in demand for our model to capture, while smoothing day-

to-day noise. Finally, the product table was joined to the aggregated dataset, and products with no designated season were removed, as our framework is designed around seasonal product groupings.

The resulting dataset contains [REDACTED] observations across [REDACTED] products. A full description of all variables is provided in Appendix B.

Finally, data were partitioned into training and testing sets by unique product identifier, with 80% of products used for model fitting and the remaining 20% for evaluating predictive accuracy on products unseen by our model. Note that products in the test set are designated as seen or unseen based on whether their product identifier appears in the training data, with all test products assumed to belong to a category present in training.

2.2 Bayesian Model Specification

We fit three Bayesian hierarchical models of varying combinations of random effects on the training dataset using the INLA package in R, which uses Integrated Nested Laplace Approximation, a deterministic and computationally efficient alternative to Markov Chain Monte Carlo (MCMC), for Bayesian inference [3]. Each of these models includes an intercept β_0 , a fixed effect β_1 for the natural logarithm of sales price, and fixed effects $\beta_2^\top$ for the remaining covariates $\mathbf{x}_{it}$. These covariates $\mathbf{x}_{it}$, each of which is described in Appendix B, include each of the binary special event flags as well as the season identifier, included to reduce confounding in the price elasticity estimate.

All three models use a log-log specification, where both quantity sold and sales price enter the model as natural logarithms. This allows β_1 to be interpreted as the population-level baseline price elasticity shared across all products, such that a 1% increase in the sales price of product i in month-year t is associated with a $\beta_1\%$ change in monthly quantity sold [4].

Let y_{it} denote the quantity sold of product i at month-year t , with expected value $\mu_{it} = \mathbb{E}[y_{it}]$. The linear predictor, excluding random effects, is given by Equation 1:

$$\log(\mu_{it}) = \beta_0 + \beta_1 \log(\text{price}_{it}) + \beta_2^\top \mathbf{x}_{it} \quad (1)$$

2.2.1 Observation Model

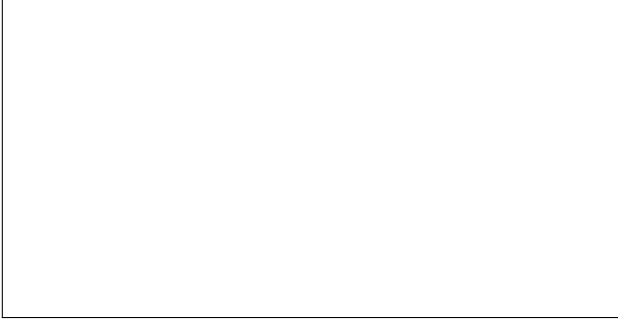


Figure 1: (Redacted) Histogram of observed quantity sold by product, month, and year (truncated at 100 for visual clarity) overlaid in green by a negative binomial distribution with mean and size parameters estimated respectively as $\hat{\mu} = \text{[redacted]}$, $\hat{r} = \text{[redacted]}$

Quantity sold is strictly positive, discrete, and heavily right-skewed, as illustrated in Figure 1. These characteristics are consistent with both the Poisson and negative binomial distributions [5]. However, the Poisson assumption of equal mean and variance is clearly violated as the sample mean of quantity sold is [redacted] and the sample variance is [redacted], indicating strong overdispersion. Additionally, as depicted in Figure 1, the negative binomial distribution fits our observed quantity sold well. We therefore model quantity sold using a negative binomial distribution.

Let $r > 0$ represent the size (dispersion) parameter of the negative binomial distribution, where smaller values of r indicate greater overdispersion relative to the Poisson distribution. Following [5], the negative binomial log-likelihood is given by:

$$\ell(\mu_{it}, r) = \sum_{i,t} [\log \Gamma(r + y_{it}) - \log \Gamma(r) + y_{it} \log \left(\frac{\mu_{it}}{\mu_{it} + r} \right) + r \log \left(\frac{r}{\mu_{it} + r} \right)] \quad (2)$$

where $\hat{\mu} = \bar{y}$ produces a closed-form solution, and $\hat{r}$ is obtained via numerical optimisation of Equation 2, producing the observation model:

$$y_{it} \sim \text{NegBin}(\mu_{it}, r) \quad (3)$$

We note a limitation of this model is that the negative binomial distribution has support on $\{0, 1, 2, \dots\}$, but zero quantities sold are unobserved in our data. Because we are sampling conditional on $y_{it} \geq 1$, estimates of μ_{it} and r may be biased upward.

2.2.2 Random Effects Model

Table 1: Summary of models fitted, differing in the random effects included.

Model	Group intercept (u_{0g})	Group slope (u_{1g})	Product intercept (v_{0i})	Product slope (v_{1i})
M1				
M2	✓	✓		
M3	✓	✓	✓	✓

In this study, three models were compared, each of which differs in terms of their product-related random effects. By doing this, we evaluate whether accounting for heterogeneity in baseline demand and price elasticity across products improves estimation of price elasticity.

As outlined in Table 1, the first model M1 contains no random effects, making it identical to the baseline in Equation 1. This model pools a single elasticity estimate across all products, ignoring product-related heterogeneity in both baseline demand and price elasticity and serving as a benchmark against which we can compare the hierarchical models.

The second model, M2, outlined in Equation 4, includes both a random intercept $u_{0g(i)}$ and a random slope $u_{1g(i)}$ at the main product category level, where $g(i)$ denotes the category to which product i belongs, allowing baseline demand and price elasticity to vary across product groups.

$$\log(\mu_{it}) = \beta_0 + \beta_1 \log(\text{price}_{it}) + \beta_2^\top \mathbf{x}_{it} + u_{0g(i)} + u_{1g(i)} \cdot \log(\text{price}_{it}) \quad (4)$$

The third model, M3, defined in Equation 5, extends M2 by additionally including a random intercept v_{0i} and random slope v_{1i} at the product identifier level, which allows each product to have its own baseline demand and price elasticity estimate. Through this structure, products borrow information from their own sales history and similar products within the same category, partially pooling toward the group-level estimates u_{0g} and u_{1g} . This seeks to address unreliable elasticity estimates for products with limited but existing sales history.

$$\log(\mu_{it}) = \beta_0 + \beta_1 \log(\text{price}_{it}) + \beta_2^\top \mathbf{x}_{it} + u_{0g(i)} + u_{1g(i)} \cdot \log(\text{price}_{it}) + v_{0i} + v_{1i} \cdot \log(\text{price}_{it}) \quad (5)$$

Note that for both M2 and M3, the main product category (as defined in Appendix B) was used as the grouping variable for category-level random effects. Because our goal is to estimate price elasticities across products, we choose a grouping variable that captures product characteristics instead of customer or temporal attributes, which makes product type (main, sub and detail categories) the most natural choice. Sub-category and detail category

were not used due to high category imbalance and computational cost. However, exploring other grouping granularities could be an extension to this work.

The random intercepts u_{0g} , v_{0i} and random slopes u_{1g} , v_{1i} are each modelled as independent and identically distributed (IID) Gaussian random effects,

$$u_{0g} \sim \mathcal{N}(0, \sigma_{u_0}^2), \quad u_{1g} \sim \mathcal{N}(0, \sigma_{u_1}^2), \quad (6)$$

$$v_{0i} \sim \mathcal{N}(0, \sigma_{v_0}^2), \quad v_{1i} \sim \mathcal{N}(0, \sigma_{v_1}^2), \quad (7)$$

where $\sigma_{u_0}^2$, $\sigma_{u_1}^2$, $\sigma_{v_0}^2$, and $\sigma_{v_1}^2$ are the corresponding variances. The IID assumption was made as we do not have any prior reason to expect baseline demand correlates with price elasticity at either the group or product-level [6].

The dependencies between model parameters, latent effects, and observed data for our most complex model, M3, are visualised in Figure 2 as a directed acyclic graph (DAG). The simpler models, M1 and M2, are nested versions of this structure.

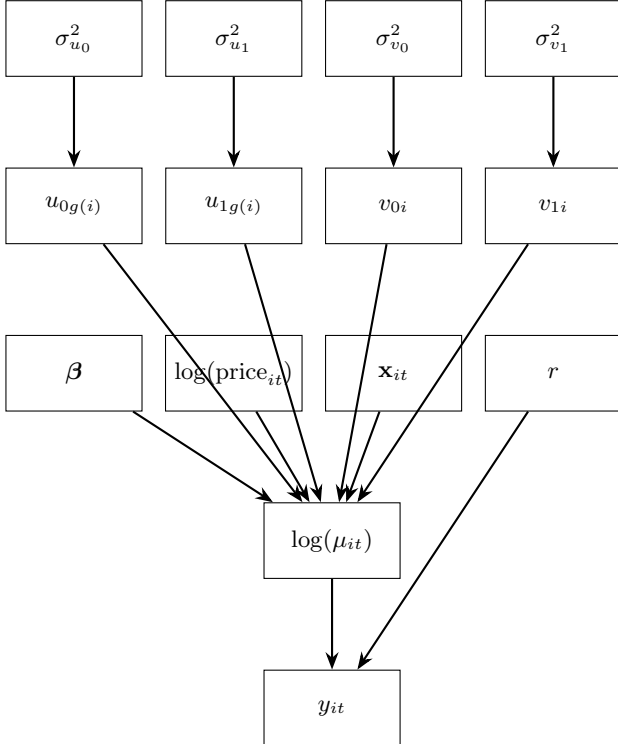


Figure 2: Directed acyclic graph (DAG) of the hierarchical Bayesian model M3. $\log(\text{price}_{it})$, though also an observed covariate, is shown separately from $\mathbf{x}_{it}$ to highlight its interaction with the group and product-level random slopes, as depicted in Equation 5.

2.2.3 Prior Model

According to the law of demand, holding all else constant, increases in price are expected to reduce quantity demanded, implying a negative relationship between price and demand [7]. Therefore, we assign a weakly informative

prior on β_1 ,

$$\beta_1 \sim \mathcal{N}(-1, 1) \quad (8)$$

This reflects the expectation of a negative effect of price on quantity sold, with a variance of 1 on the log-log scale, ruling out implausible elasticity values while still allowing a wide enough range to capture realistic values.

As an evaluation of population-level elasticity robustness, we conducted a sensitivity analysis on β_1 under three prior specifications: the base prior $\mathcal{N}(-1, 1)$, a vague prior $\mathcal{N}(0, 1000)$, and a strong prior $\mathcal{N}(-1, 0.25)$, fitted under M3 with results presented in Appendix C. While the vague prior produces a slightly narrower posterior, as it places no directional pull on the estimate, all three specifications are centred around $\beta_1 \approx \blacksquare$, suggesting elasticity estimates are robust to prior choice.

All remaining priors are set to R-INLA’s default vague priors, reflecting minimal prior knowledge and ensuring inference is driven by the data. Therefore, the following priors were specified for the fixed effects,

$$\beta_0 \sim \mathcal{N}(0, 1000), \quad \beta_2 \sim \mathcal{N}(\mathbf{0}, 1000\mathbf{I}), \quad (9)$$

These priors are centred at zero, implying no prior directional preference, and the large variance of 1000 ensures they are sufficiently vague.

Additionally, the size parameter r is assigned a penalised complexity (PC) prior [8] on $\theta = -\log(r)$, with rate parameter $\lambda = 7$. Though $\lambda = 7$ pushes the model moderately toward the Poisson distribution, the strong overdispersion observed in the data ($\hat{r} = \blacksquare$) ensures the likelihood dominates this prior.

The priors on the IID Gaussian variances $\sigma_{u_0}^2$, $\sigma_{u_1}^2$, $\sigma_{v_0}^2$, and $\sigma_{v_1}^2$ each follow a log-Gamma prior on the log-precision $\log(1/\sigma^2)$,

$$\log(1/\sigma^2) \sim \text{LogGamma}(1, 0.00005), \quad (10)$$

where the near-zero rate parameter ensures little shrinkage, allowing the data to strongly inform between-group and between-product variability.

3 Results

3.1 Model Comparison

3.1.1 In-Sample Model Selection Criteria

We assess the goodness-of-fit of M1–M3 using three scoring rules: Widely Applicable Information Criterion (WAIC) [9], which penalises model complexity; Negative Logarithm Sum of Conditional Predictive Ordinates (NLSPO) [10], which evaluates the likelihood of each observation given all remaining data points; and Marginal

[REDACTED]

As a further test of calibration, we then calculated randomised quantile residuals (RQR) by applying a normal quantile transform to the PIT values, such that a perfect calibration would produce residuals following a standard normal distribution [14]. By plotting these residuals against theoretical quantiles in the form of a quantile-quantile (QQ) plot, we can assess the calibration of different parts of the predictive distribution (see Figure 4).

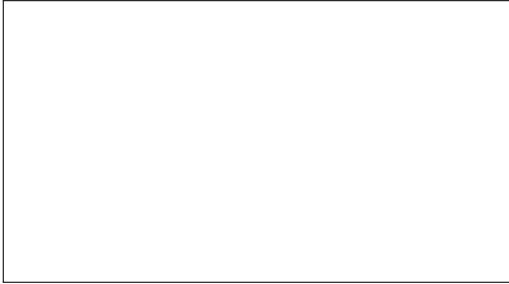


Figure 4: (Redacted) Normal QQ-plot of randomised quantile residuals across all training observations under model [REDACTED].

The residuals [REDACTED]

3.2.2 Posterior Predictive Checks

We now draw 500 replicate datasets from the fitted model’s posterior predictive distribution and compare the mean, maximum, and skewness of simulated quantities sold against the observed training data, assessing the model’s ability to capture average demand, extreme values, and distributional asymmetry respectively. We exclude the minimum from this assessment, as the negative binomial distribution generated counts such that all 500 replicates produced a minimum of zero, making it an uninformative diagnostic. These posterior predictive checks are visualised in Figure 5.

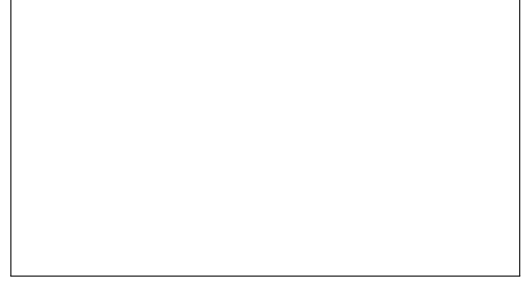


Figure 5: (Redacted) Posterior predictive histograms of 500 simulated replicates of quantity sold under model [REDACTED], with the observed summary statistic marked in red: (a) maximum, (b) mean, (c) skewness.

The most notable misfit is the mean (Plot (b)), where

[REDACTED] This is consistent with Jensen’s inequality, meaning that because the exponential function is convex, back-transforming the posterior mean on the log scale systematically underestimates the mean on the count scale, which is further amplified by the skewness of our quantities sold distribution [15]. A similar but less severe pattern is observed for the maximum (Plot (a)), where [REDACTED]. Plot (c) shows that the observed skewness is [REDACTED].

Therefore, [REDACTED]

3.2.3 Out-of-Sample Quantity Sold Diagnostics

While Section 3.1.2 used CRPS to compare models, we now examine [REDACTED] predictive behaviour on unseen products in greater detail, assessing calibration, posterior predictive intervals, and CRPS by product category.

As a complement to the in-sample calibration assessment of Section 3.2.1, we assess out-of-sample calibration by computing coverage of M3’s 95% posterior predictive intervals on the test set, generated from 500 posterior predictive samples. Coverage measures the proportion of observed values from the test set falling within the predicted interval, with a well-calibrated model achieving a 95% coverage. Our model produced a coverage of [REDACTED], indicating that our model [REDACTED], consistent with Section 3.2.1.

Given that a key aspect of our model is its use of a category- level random effects, we examine how uncertainty varies across categories by visualising posterior predictive intervals for one representative product from each

main category, selected as the product whose CRPS is closest to the median CRPS within each group (see Figure 6).

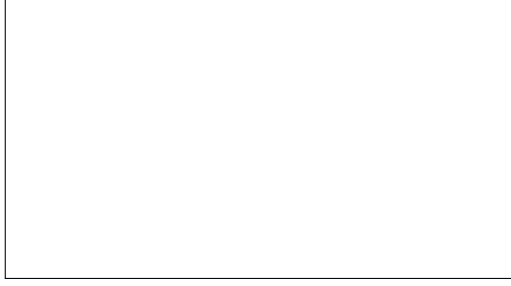


Figure 6: (Redacted) 95% posterior predictive intervals of model [REDACTED] for one product from each category on the test set, with observations ordered by increasing median predicted quantity sold. The shaded region denotes the 95% posterior predictive interval, the blue line the median prediction, and the red points the observed quantity sold.

Coverage across all four categories is [REDACTED].

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

To quantify predictive performance across all test observations in each product category, we compute individual observation-level CRPS scores and group them by category, as displayed in Figure 7.



Figure 7: (Redacted) Boxplots of observation-level CRPS values for each product category in the test set under model [REDACTED], displayed on a log scale for ease of visualisation.

As shown in Figure 7, [REDACTED].

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. This behavior may be due to the high heterogeneity in demand within this category, as it is broader,

covering a wider range of products than the others, and elasticity estimates therefore [REDACTED].

3.3 Population-Level Price Elasticity and Covariate Effects

In the summary output of [REDACTED] (see Table 4), the population-level elasticity estimate $\hat{\beta}_1 = [REDACTED]$, meaning, as discussed in Section 2.2, a 1% increase in price is associated with a [REDACTED] in monthly quantity sold. The 95% CI of [REDACTED] quantifies the posterior uncertainty around this estimate, indicating [REDACTED].

Note that this population-level estimate does not include product-specific deviations in price elasticity captured by the random effects, which will be covered in Section 3.4.2.

Table 4: Posterior summary of fixed effects for model [REDACTED]

Parameter	Mean	95% CI
Intercept	[REDACTED]	[REDACTED]
log(price)	[REDACTED]	[REDACTED]
Ramadan	[REDACTED]	[REDACTED]
Sacrifice	[REDACTED]	[REDACTED]
New Year	[REDACTED]	[REDACTED]
Mother's Day	[REDACTED]	[REDACTED]
Father's Day	[REDACTED]	[REDACTED]
Valentine's Day	[REDACTED]	[REDACTED]
Singles' Day	[REDACTED]	[REDACTED]
Black Friday	[REDACTED]	[REDACTED]
Gender: Unisex	[REDACTED]	[REDACTED]
Gender: Women	[REDACTED]	[REDACTED]
Season: Spring-Summer	[REDACTED]	[REDACTED]

The remaining fixed effects of Table 4 tell us how each covariate influences demand. Notably, [REDACTED].

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

The 95% CIs of these covariates quantify posterior uncertainty around each estimate, [REDACTED].

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

We additionally interpret the intercept estimate of $\hat{\beta}_0 = [REDACTED]$ as the expected log quantity sold when the log price equals zero (i.e., a price of 1 unit), all special event

indicators equals zero (i.e., no special event occurs), the product gender is male, and the product is designed for the fall-winter season.

3.4 Heterogeneity in Price Elasticities and Prediction

We now discuss how price elasticity varies across individual products and product categories through the random effects of β_i . We will then explore how these estimates combine with the population-level estimate $\hat{\beta}_1$ to yield product-specific predictions of price elasticity.

3.4.1 Heterogeneity in Price Elasticity

Table 5 displays the summary output of the category-level random effects of β_i , including estimates for the intercept $\hat{u}_{0g}$ and slope $\hat{u}_{1g}$ of Equation 5.

Table 5: Posterior summary of category-level random effects for model β_i

Category	Intercept ($\hat{u}_{0g}$)		Slope ($\hat{u}_{1g}$)	
	Mean	95% CI	Mean	95% CI
Accessories	0.00	[-0.05, 0.05]	0.00	[-0.05, 0.05]
Bags	0.00	[-0.05, 0.05]	0.00	[-0.05, 0.05]
Cosmetics	0.00	[-0.05, 0.05]	0.00	[-0.05, 0.05]
Ready-to-Wear	0.00	[-0.05, 0.05]	0.00	[-0.05, 0.05]
Shoes	0.00	[-0.05, 0.05]	0.00	[-0.05, 0.05]

Category-level intercept estimates $\hat{u}_{0g}$ represent deviations in baseline demand from the population average $\hat{\beta}_0 = 0.00$. As we can see, $\hat{u}_{0g}$ is centered around zero for all categories.

The slope estimates $\hat{u}_{1g}$ capture category-level deviations in price elasticity from $\hat{\beta}_1 = 0.00$. As shown, $\hat{u}_{1g}$ is centered around zero for all categories.

The slope estimates $\hat{u}_{1g}$ capture category-level deviations in price elasticity from $\hat{\beta}_1 = 0.00$. As shown, $\hat{u}_{1g}$ is centered around zero for all categories.

With regard to the uncertainty around $\hat{u}_{0g}$, $\hat{u}_{1g}$ is centered around zero for all categories.

With regard to the uncertainty around $\hat{u}_{0g}$, $\hat{u}_{1g}$ is centered around zero for all categories.

We now evaluate the product-level random effects of β_i , including estimates for the intercept $\hat{v}_{0i}$ and slope $\hat{v}_{1i}$ of Equation 5. Given the large number of unique product identifiers, we summarise the posterior product-level estimates using a caterpillar plot, where black points denote the posterior mean estimates and horizontal lines represent 95% CIs, ordered by posterior mean (see Figure 8).



Figure 8: (Redacted) Caterpillar plot of 100 randomly selected product-level random effects (subsampled for visual clarity), sorted by posterior mean, with a red reference line at zero. (a) Random intercepts $\hat{v}_{0i}$, representing product-specific deviations from $\hat{\beta}_0$. (b) Random slopes $\hat{v}_{1i}$, representing product-specific deviations in price elasticity from $\hat{\beta}_1$.

As visualised in Plot (a), the random intercept estimates $\hat{v}_{0i}$ span from approximately -0.05 to 0.05 , indicating $\hat{v}_{0i}$ is centered around zero for all products.

The range of approximately -0.05 to 0.05 in Plot (b) suggests $\hat{v}_{1i}$ is centered around zero for all products.

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3.4.2 Elasticity Estimation and Prediction via Hierarchical Random Effects

We estimate the population-level baseline price elasticity with $\hat{\beta}_1$, however, to incorporate both the product and category-level random effects information, we calculate the elasticity estimate $\hat{\epsilon}_i$ as,

$$\hat{\epsilon}_i = \hat{\beta}_1 + \hat{u}_{1g(i)} + \hat{v}_{1i} \quad (11)$$

For unseen products with no sales history, $\hat{v}_{1i}$ is unavailable, and to predict price elasticity we rely solely on $\hat{\beta}_1 + \hat{u}_{1g(i)}$, pooling toward the category mean.

We then quantify uncertainty around this estimate by drawing 500 posterior samples of $\hat{\beta}_1$, $\hat{u}_{1g(i)}$, and $\hat{v}_{1i}$ where applicable, and taking the 2.5th and 97.5th percentiles of their sum to produce a 95% CI.

Table 6 illustrates this process for one randomly selected seen and one unseen product. As no ground truth elasticity exists, these examples demonstrate the predic-

tion mechanism instead of validating it; model validation was instead addressed in Sections 3.1 and 3.2.

Table 6: Price elasticity estimation for one seen product and prediction for one unseen product, with 95% credible intervals.

	Product ID	$\hat{\beta}_1$	$\hat{u}_{1g(i)}$	$\hat{v}_{1i}$	$\hat{\varepsilon}_i$	95% CI
Seen	P...16056					
Unseen	P...00970			NA		

Consistent with Section 3.4.1,

4 Discussion

4.1 Limitations

While , some limitations warrant consideration. Firstly, our negative binomial model has support on $\{0, 1, 2, \dots\}$, but the minimum observable quantity sold is 1. This biases our estimates of μ and r upward, suggesting our elasticity estimates may be biased toward zero. A zero-truncated negative binomial model would address this directly [16], but is not natively supported in R-INLA and is therefore left as a future direction.

Secondly, we evaluated predictive accuracy of quantity sold, but lacked ground-truth elasticities to validate our estimates directly. Though we did investigate the robustness of our population-level elasticity estimate via sensitivity analysis (Appendix C), future work could obtain ground-truth elasticities via controlled experiments or simulation studies.

The dispersion and tail behavior of our model

A generalised negative binomial distribution, which has an additional dispersion parameter, may better capture this behaviour [17], but is also not natively supported in R-INLA.

Our model also

due to Jensen’s inequality. Exploring alternative link functions or data transformations could address this in future work.

Lastly, when assessing our model’s out-of-sample

predictive ability,

. Investigating alternative product groupings may improve predictive performance in future work.

4.2 Conclusion

The goal of this report was to estimate and predict price elasticities for newly launched fast fashion products with limited sales history, assessing the value of product and category-level random effects in capturing heterogeneous price sensitivities, and to quantify uncertainty around these estimates.

After fitting three log-log negative binomial Bayesian hierarchical models of varying complexities, we found that

. Using this selected model, we found that

, with a population-level elasticity of $\hat{\beta}_1 =$, meaning a 1% increase in price is associated with a in monthly quantity sold.

Product-specific elasticities were estimated by summing the population-level elasticity with category and product-level random slope deviations ($\hat{\varepsilon}_i = \hat{\beta}_1 + \hat{u}_{1g(i)} + \hat{v}_{1i}$) for seen products, with uncertainty quantified via 95% credible intervals, with predictions on unseen products excluding $\hat{v}_{1i}$.

When evaluating predictive accuracy on unseen products,

We therefore recommend that Simon-Kucher incorporate

into elasticity estimation for pricing decisions on products with existing sales histories. For newly launched products with no sales history,

. In either case,

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Appendices

A Special Event Flags

Table 7: Daily event flags used in preprocessing.

Event	Component Flags
Ramadan	First 10 days (2023), First 10 days (2024–2025), 3 days before, Eve, Eid day 1, Remaining Eid days
Eid al-Adha	3 days before, Eve, Eid day 1, Remaining Eid days
New Year	7 days before, 3 days before, New Year’s Day
Mother’s Day	2 days before, Mother’s Day
Father’s Day	2 days before, Father’s Day
Valentine’s Day	2 days before, Valentine’s Day
Singles’ Day	1 day before, Singles’ Day
Black Friday	Black Friday

B Dataset Variable Definitions

Table 8: Definitions of variables used after preprocessing.

Variable	Type	Description
Product identifier	Character	Unique product identifier
Month	Integer	Month of observation (1–12)
Year	Integer	Year of observation (2023–2025)
Quantity sold	Integer	Total units sold
Sales price	Numeric	Median sales price
Ramadan indicator	Binary	1 if the month contains a Ramadan event (see Appendix A)
Eid al-Adha indicator	Binary	1 if the month contains an Eid al-Adha event (see Appendix A)
New Year indicator	Binary	1 if the month contains a New Year event (see Appendix A)
Mother’s Day indicator	Binary	1 if the month contains a Mother’s Day event (see Appendix A)
Father’s Day indicator	Binary	1 if the month contains a Father’s Day event (see Appendix A)
Valentine’s Day indicator	Binary	1 if the month contains a Valentine’s Day event (see Appendix A)
Singles’ Day indicator	Binary	1 if the month contains a Singles’ Day event (see Appendix A)
Black Friday indicator	Binary	1 if the month contains Black Friday (see Appendix A)
Main category	Character	Top-level product category
Sub-category	Character	Mid-level product category
Detail category	Character	Fine-grained product category
Season identifier	Character	Season the product was designed for

C Sensitivity Analysis of Population-Level Price Elasticity



Figure 9: (Redacted) Posterior density of β_1 under three prior specifications for model M1

D Category-Level Slope Comparison: Model M2 vs M3

Table 9: Category-level slope ($\hat{u}_{1g}$) 95% credible intervals and widths for models M2 and M3.

Category	M2 Slope ($\hat{u}_{1g}$)		M3 Slope ($\hat{u}_{1g}$)	
	95% CI	Width	95% CI	Width
Accessories				
Bags				
Cosmetics				
Ready-to-Wear				
Shoes				